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DETAIL SPECIFICATION

VALVE, HOSE ASSEMBLY, AND ADAPTER, VENT AND TEST HYDRAULIC SERVICE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vent and test valves, hose assemblies, and adapters for use in hydraulic service to a maximum operating pressure of 310 bars [4,500 pounds per square inch (lb/in^2)].

1.2 Classification. Vent and test valves, hose assemblies, and adapters are furnished in the configuration and sizes listed in the applicable specification sheet (see 3.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service

MIL-PRF-17672 - Hydraulic Fluid, Petroleum, Inhibited

MIL-H-19457 - Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-H-22072 - Hydraulic Fluid, Catapult, NATO Code Number H-579

MIL-DTL-24695/1 - Valve, Vent and Test Hydraulic Service

MIL-DTL-24695/2 - Vent and Test Valve Hose Assembly Hydraulic Service

MIL-DTL-24695/3 - Vent and Test, Hose Assembly Adapter

MIL-DTL-32353 - Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibration of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This/these document/s is/are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS-QQ-S-763 - Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

SAE AS3581 - Packing, Preformed - O-Ring Seal AMS7259

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Vent and test valves, hose assemblies, and adapters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Materials.

3.3.1 Compatibility. Where specified herein, parts shall be constructed of materials that are compatible with the following hydraulic service fluids: MIL-PRF-17672, MIL-H-19457, MIL-H-22072, MIL-PRF-17331, and MIL-DTL-32353 (see 3.3.6 and 3.3.7). Verification of material compatibility shall be by analysis or test (see 4.2.6).

3.3.2 Prohibited materials. Vent and test valves, hose assemblies, and adapters shall not use the following materials:

a. Chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

b. Zinc or zinc plated materials.

c. Mercury or mercury compounds.

d. Magnesium or magnesium base alloys.

e. Radioactive materials.

f. Cadmium plated materials.

g. Beryllium or beryllium compounds.

3.3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.4 Nickel alloys. Nickel alloys for use in pressure-containing vent and test valve, hose assembly, or adapter parts shall be limited to nickel-copper (NiCu) alloy: QQ-N-281, Class A (see 4.2.3.1, 4.2.4.1, and 4.2.5.1).

3.3.5 Corrosion resistant steel (CRES). The use of chromium-nickel austenitic steel in pressure-containing vent and test valve, hose assembly, or adapter parts shall be limited to Class 304, 304L, 316, or 316L in accordance with SAE AMS-QQ-S-763; Type 316Ti in accordance with ASTM A276/A276M; or other steel as approved by the qualifying activity (see 4.2.3.1, 4.2.4.1, and 4.2.5.1).

3.3.6 Non-metallic materials. Vent and test valve cap, if fabricated from non-metallic materials, shall be compatible with all hydraulic service fluids specified herein (see 3.3.1). Hose assembly parts fabricated from non-metallic materials shall be compatible with all hydraulic service fluids specified herein (see 3.3.1).

3.3.7 O-rings. Internal O-ring seals shall be compatible with all hydraulic service fluids specified herein (see 3.3.1) or shall be in accordance with SAE AS3581 (see 4.2.3.1 and 4.2.5.1). External O-ring seals shall be fluorocarbon elastomer in accordance with SAE AS3581 and shall be provided with the valve (see 4.2.3.1).

3.4 Construction.

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3.4.1 Vent and test valves. Vent and test valves shall be constructed as specified herein and in the applicable specification sheet. The vent and test valves furnished under this specification shall seal by means of a spring loaded check valve. A probe, integral to the end fitting of the hose assembly and conforming to the configuration identified in the applicable specification sheet, shall be used to open the check valve by depressing the ball or poppet, opening the flow passage. When specified by the specification sheet (see MIL-DTL-24695/1) part number designation, a lock wire hole on the hex portion of the valve shall be provided in accordance with the specification sheet.

3.4.1.1 Strength. Vent and test valves shall withstand the structural loads imposed by the test requirements of this specification. Valves shall withstand the wrench loads required for installation. The use of aluminum alloys for threaded applications is prohibited.

3.4.1.2 Clearance. Within the most adverse dimensions, there shall be clearance of moving parts at 0 and at 70 °C (32 and at 158 °F). The room temperature reference point shall be 20 °C (68 °F).

3.4.1.3 Cap design. A knurled valve cap shall be attached to the valve for installation when the hose assembly is not connected. The cap shall be equipped with a male connection to mate with the probe seal in the valve. This male connection shall not be long enough to actuate the valve. This male connection and the probe seal shall form a seal that prevents leakage at 310 bars (4,500 lb/in^2) (see 4.2.3.5 and 4.3.1.2.1.3).

3.4.2 Hose assemblies. Hose assemblies shall be constructed as specified herein and in the applicable specification sheet. The hose assembly shall have a probe which is integral to the reverse buttress thread end fitting. An O-ring seal, located within the vent and test valve, shall seal against the probe (see 3.4.1). Dimensions shall conform to the applicable specification sheet. The probe shall be used to depress the check valve ball or poppet, opening the flow passage. The minimum hose internal diameter shall be not less than 1.0 millimeter (0.039 inch).

3.4.3 Adapters. Adapters shall be constructed as specified herein and in the applicable specification sheet. Dimensions shall conform to the applicable specification sheet. Adapters shall be equipped with a locking elastomer ring to dampen shock and vibration that may loosen the adapter. The adapter shall be sized to mate with the vent valve hose fitting (see MIL-DTL-24695/2 for detail of vent and test valve hose assembly end fitting).

3.5 Performance. Vent and test valves, hose assemblies, and adapters shall perform as follows (see 4.2).

3.5.1 Vent and test valves.

3.5.1.1 Proof pressure. The valve shall withstand a proof pressure of 465 bars (6,750 lb/in^2) without signs of external leakage, failure, or permanent set (see 4.2.3.2). The valve cap shall withstand a proof pressure of 465 bars (6,750 lbs/in^2) without signs of external leakage, failure, or permanent set (see 4.2.3.2).

3.5.1.2 Shock. The valve shall meet the shock requirements as specified in MIL-DTL-901 for Grade A, Class I equipment (see 4.2.3.3).

3.5.1.3 Leakage. Leakage shall be limited to those criteria specified in the leakage tests with the valve uncapped and capped (see 4.2.3.4 and 4.2.3.5).

3.5.1.4 Operation. The torque required to couple the hose assembly to the valve under the maximum operating pressure of 310 bars (4,500 lb/in^2) shall not exceed 2.0 Newton meters (18 inch-pounds) (see 4.2.3.6).

3.5.1.5 Endurance. The valves shall withstand 250 open/shut cycles or seating/unseating of the valve's internal check or poppet by the hose assembly probe or equivalent substitute at an operating pressure of 310 bars (4,500 lb/in^2) at room temperature without evidence of external leakage to exceed one drop during a 2-hour period following valve cycling (see 4.2.3.6).

3.5.1.6 Vibration. The valves shall operate as specified herein after being subjected to vibration tests in accordance with MIL-STD-167-1 (see 4.2.3.7). Presence of any of the following conditions resulting from vibration shall be cause for rejection:

a. Damage to parts.

b. Loosening of parts (including cap).

c. Leakage of one drop or more from the uncapped valve.

d. Loosening (unscrewing) of the valve when the cap is removed.

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3.5.1.7 Burst pressure. The valve shall withstand a burst pressure of 620 bars (9,000 lb/in^2) without any sign of rupture or permanent deformation (see 4.2.3.8).

3.5.1.8 Valve flow rate. The vent and test valve, when connected to the hose assembly and adapter, shall pass a minimum of 250 milliliters (15.3 in^3) per minute at a pressure differential of 10 bars (145 lb/in^2) (see 4.2.3.9).

3.5.2 Hose assembly.

3.5.2.1 Proof pressure. The hose assembly shall withstand a proof pressure of 620 bars (9,000 lb/in^2) (see 4.2.4.2).

3.5.2.2 Flexibility. The hose assembly shall exhibit no evidence of permanent deformation when pressurized to 310 bars (4,500 lb/in^2) operating pressure at a bend radius of 40 millimeters (1.57 inches) (see 4.2.4.3).

3.5.2.3 Leakage. The hose assembly shall show no sign of leakage after 2 hours under an operating pressure of 310 bars (4,500 lb/in^2) (see 4.2.4.4).

3.5.2.4 Burst pressure. The hose assembly shall withstand a burst pressure of 1,240 bars (18,000 lb/in^2) with no sign of leakage (see 4.2.4.5).

3.5.3 Adapters.

3.5.3.1 Proof pressure. The adapter shall withstand a proof pressure of 465 bars (6,750 lb/in^2) without permanent deformation or damage (see 4.2.5.2).

3.5.3.2 Operation. The torque required to couple hose assembly to the adapter shall not exceed 2.0 Newton meters (18 inch-pounds) (see 4.2.5.4).

3.5.3.3 Leakage. The adapter shall show no sign of leakage after 2 hours under an operating pressure of 310 bars (4,500 lb/in^2) (see 4.2.5.3).

3.5.3.4 Burst pressure. The adapter shall withstand a burst pressure of 620 bars (9,000 lb/in^2) without any sign of rupture or permanent deformation (see 4.2.5.5).

3.6 Marking. Each valve and each hose assembly shall be identified with a metal or plastic tag (see 4.2.3.1 and 4.2.4.1) bearing the following information:

a. Manufacturer’s name, trademark, or logo.

b. Specification sheet part number.

c. Manufacturer’s part number.

d. Assembly date (by quarter and year).

3.7 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength (see 4.2.3.1).

3.8 Workmanship. Vent and test valves, hose assemblies, and adapters shall be free from porosity, roughness, or any other defect which might affect their serviceability (see 4.2.3.1, 4.2.4.1, and 4.2.5.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of the tests and examinations as specified in 4.2.1 through 4.2.7 and shall be conducted in the order as shown in tables I, II, and III unless otherwise approved by the qualifying activity.

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TABLE I. Valve qualification inspection.

Inspection	Order of tests	Requirement	Test
Examination	1	3.3.2, 3.3.4, 3.3.5, 3.3.7, 3.4.1, 3.6, 3.7, 3.8	4.2.3.1, 4.2.7
Compatibility	2	3.3.1	4.2.6
Shock	3	3.5.1.2	4.2.3.3
Vibration	4	3.5.1.6	4.2.3.7
Assembly flow rate	5	3.5.1.8	4.2.3.9
Proof pressure	6	3.5.1.1	4.2.3.2
Leakage – uncapped	7	3.5.1.3	4.2.3.4
Leakage – capped	8	3.5.1.3	4.2.3.5
Operation	9	3.5.1.4	4.2.3.6
Endurance	10	3.5.1.5	4.2.3.6
Burst pressure	11	3.5.1.7	4.2.3.8

TABLE II. Hose assembly qualification inspection.

Inspection	Order of tests	Requirement	Test
Examination	1	3.3.2, 3.3.4, 3.3.5, 3.4.2, 3.6, 3.8	4.2.4.1, 4.2.7
Compatibility	2	3.3.1	4.2.6
Proof pressure	3	3.5.2.1	4.2.4.2
Flexibility	4	3.5.2.2	4.2.4.3
Leakage	5	3.5.2.3	4.2.4.4
Burst pressure	6	3.5.2.4	4.2.4.5

TABLE III. Adapter qualification inspection.

Inspection	Order of tests	Requirement	Test
Examination	1	3.3.2, 3.3.4, 3.3.5, 3.3.7, 3.4.3, 3.8	4.2.5.1, 4.2.7
Proof pressure	2	3.5.3.1	4.2.5.2
Operation	3	3.5.3.2	4.2.5.4
Leakage	4	3.5.3.3	4.2.5.3
Burst pressure	5	3.5.3.4	4.2.5.5

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4.2.1 Samples for qualification. Samples for the qualification inspection shall consist of one vent and test valve, one adapter, and two hose assemblies for each part number as specified on the applicable specification sheet.

4.2.1.1 Extension of qualification. Qualification of one hose assembly length can be extended to all lengths provided that the end fittings are identical for each assembly and only the hose length varies. For valves and adapters whose only difference is the end fitting configuration, qualification can be extended provided a burst pressure test is conducted.

4.2.2 Test conditions. Unless otherwise specified herein, the hydraulic fluid used for all tests shall conform to MIL-PRF-17672, 2135T-H. The fluid shall be maintained at a minimum temperature of 45 °C (113 °F).

4.2.3 Vent and test valves.

4.2.3.1 Examination. Each valve shall be examined for conformance to the requirements of this specification, applicable specification sheets, and manufacturer's drawings.

4.2.3.2 Proof pressure (uncapped valve and capped valve).

a. Proof pressure (uncapped valve). The uncapped valve shall withstand a proof pressure of 465 bars (6,750 lb/in^2) applied to the inlet with the outlet port blocked without evidence of permanent deformation, malfunction, or leakage other than a slight wetting at seals. Proof pressure shall be applied at least two successive times and held 2 minutes for each pressure application. The valve shall be operated between pressure applications. The pressure shall be reduced to atmospheric pressure between applications.

b. Proof pressure (capped valve). The male connection of the valve cap and the probe seal in the valve shall withstand a proof pressure of 465 bars (6,750 lb/in^2) without evidence of failure, permanent set, or leakage at the seal other than a slight wetting insufficient to form a drop. Proof pressure shall be applied once and held for 1 minute. This test may be accomplished on a separate valve with the check valve device removed.

4.2.3.3 Shock. The valve shall be subjected to high-impact mechanical shock tests in accordance with the requirements for Grade A, subsidiary component, Class I, lightweight, Type B, fixture 4A of MIL-DTL-901. A valve shall be considered as having failed the shock test if it does not meet the leakage test of 4.2.3.4, or shows any signs of external or internal damage when examined in accordance with 4.2.3.1. The valve shall be pressurized to the maximum operating pressure of 310 bars (4,500 lb/in^2) during the test.

4.2.3.4 Leakage (uncapped valve). The valve shall be uncapped and subjected to the operating pressure of 310 bars (4,500 lb/in^2). The valve shall then be actuated with the probe and the operating pressure maintained for 2 hours. There shall be no leakage other than a slight wetting of the seal. During connection to the probe, one drop shall be allowed. The test shall be repeated at 50 percent of the specified operating pressure and at 0.35 bars (5 lb/in^2). The valve shall automatically close after disconnection from the probe without leakage.

4.2.3.5 Leakage (capped valve). The cap shall prevent leakage at 310 bars (4,500 lb/in^2) in the event of total valve failure. This pressure shall be held for 2 hours. During this time, leakage shall be insufficient to form a drop. This test may be accomplished on a separate valve, with the check valve device removed, at any time in the testing sequence.

4.2.3.6 Valve operation. The valve inlet shall be pressurized to 310 bars (4,500 lb/in^2) and torque required to connect the hose assembly to the valve shall be measured. The torque shall not exceed the limit specified herein (see 3.5.1.4). In addition, the valve shall be pressurized to 310 bars (4,500 lb/in^2) and operated for 250 open/shut cycles. This may be accomplished by seating/unseating the valve's internal check or poppet by the hose assembly probe or equivalent substitute at room temperature. After cycling, the leakage test of 4.2.3.4 shall be performed. Leakage shall not exceed the limit specified herein (see 3.5.1.5).

4.2.3.7 Vibration. The valve shall be pressurized and subjected to Type I (environmental vibration) vibration tests in accordance with MIL-STD-167-1 for the frequency of 4 to 50 Hertz. The valve shall be tested in each of three mutually perpendicular axes. However, if a component is symmetrical about a principal axis, vibration testing need only be accomplished along two orthogonal axes, one of which shall be the principle axis of symmetry. For the last axis tested, the valve cap shall not be installed. Any of following conditions resulting from vibration shall constitute failure of this test:

a. Damage to parts.

b. Loosening of parts (including cap).

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c. External leakage of one drop or more from the uncapped valve.

d. Loosening (unscrewing) of the valve when the cap is removed [for vibration test, installation torque applied for installing valve shall be 11 to 13.5 Newton meters (99 to 121.5 inch-pounds) which is one-half the normal torque value of 22 to 27 Newton meters (198 to 243 inch-pounds)].

4.2.3.8 Burst pressure. Pressure shall be applied to the valve at a rate not to exceed 1,725 bars (25,000 lb/in^2) per minute until 620 bars (9,000 lb/in^2) burst pressure is obtained. After this pressure has been held for 2 minutes, the valve shall show no sign of rupture or permanent deformation. The pressure may be further increased to determine the actual burst pressure.

4.2.3.9 Valve flow rate. It shall be demonstrated that the vent and test valve, when connected to the hose assembly and adapter, provides a minimum flow rate of 250 milliliters (15.3 in^3) per minute at a differential pressure of 10 bars (145 lb/in^2). Minimum hose length shall be 915 millimeters (36 inches).

4.2.4 Hose assembly.

4.2.4.1 Examination. Each hose assembly shall be examined for conformance to the requirements of this specification, applicable specification sheets, and manufacturer's drawings.

4.2.4.2 Proof pressure. The hose assembly shall withstand a proof pressure of 620 bars (9,000 lb/in^2) without evidence of permanent deformation or external leakage. The proof pressure shall be applied at least twice and held for 2 minutes at each application. The pressure shall be reduced to atmospheric pressure between applications.

4.2.4.3 Flexibility. The hose assembly shall be pressurized to the 310 bars (4,500 lb/in^2) operating pressure. The assembly shall be flexed to the minimum bend radius of 40 millimeters (1.57 inches) and the pressure increased to the applicable proof pressure. Pressure shall be reduced to atmospheric pressure, and the test repeated. Any evidence of permanent deformation or leakage shall constitute failure of this test.

4.2.4.4 Leakage. The hose assembly shall be subjected to the 310 bars (4,500 lb/in^2) operating pressure with each end connected to either a MIL-DTL-24695/1 vent and test fitting or a MIL-DTL-24695/3 adapter. After 2 hours there shall be no evidence of leakage.

4.2.4.5 Burst pressure. Pressure shall be applied to the hose assembly at a rate not to exceed 1,725 bars (25,000 lb/in^2) per minute until the 1,240 bars (18,000 lb/in^2) burst pressure is obtained. After this pressure has been held for 2 minutes, the hose and end fittings shall show no sign of leakage. Following this verification, either the pressure shall be further increased to determine the actual burst pressure or other action shall be taken to render the hose assembly unfit for use.

4.2.5 Adapters.

4.2.5.1 Examination. Each adapter shall be examined for conformance to the requirements of this specification, applicable specification sheets, and manufacturer's drawings.

4.2.5.2 Proof pressure. The adapter shall withstand a proof pressure of 465 bars (6,750 lb/in^2) applied to the inlet with the outlet port blocked without evidence of permanent deformation or damage.

4.2.5.3 Leakage. The adapter shall be subjected to the 310 bars (4,500 lb/in^2) operating pressure. After 2 hours there shall be no evidence of leakage.

4.2.5.4 Adapter operation. The adapter outlet shall be pressurized to 310 bars (4,500 lb/in^2) and torque required to connect the hose assembly to the adapter shall be measured. The torque shall not exceed the limit specified herein (see 3.5.3.2).

4.2.5.5 Burst pressure. Pressure shall be applied to the adapter at a rate not to exceed 1,725 bars (25,000 lb/in^2) per minute until 620 bars (9,000 lb/in^2) burst pressure is obtained. The burst pressure shall be held for 2 minutes without any sign of rupture or permanent deformation. The pressure may be further increased to determine the actual burst pressure.

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4.2.6 Non-metallic parts compatibility. Where specified herein (see 3.3.6 and 3.3.7), non-metallic parts shall be verified for compatibility with hydraulic service fluids (see 3.3.1) by analysis or test. If verification is by test, parts shall be immersed in each of the hydraulic service fluids for 7 days at ambient pressure and temperature. Following immersion, parts shall be examined for evidence of swelling, shrinkage, or other material degradation that would adversely affect performance or operability. Any evidence of adverse material degradation by examination or subsequent performance or operability testing in the test fluid (see 4.2.2) shall be cause for rejection or non-qualification.

4.2.7 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3.2 as required by the qualifying activity.

4.3 Conformance inspection. Conformance inspection shall consist of the examination of 4.3.1.1 and the tests of 4.3.1.2. Each valve, hose assembly, and adapter shall be subjected to conformance inspection, except that testing of the hose assembly and dimensional inspection of the valve, hose assembly, and adapter may be accomplished by a sample plan using table IV, based on the batch or the lot size. Non-compliance with conformance inspection criteria shall result in rejection of the non-compliant product and a 100 percent inspection of the entire batch or a lot.

TABLE IV. Sample size for hose assembly testing.

Batch or lot size	Sample size
1	1
2-15	2
16-25	3
26-90	5
91-150	8
151-280	13
281-500	20

4.3.1 Examination and tests.

4.3.1.1 Examination.

4.3.1.1.1 Vent and test valves. Each vent and test valve shall be visually examined to determine conformance to the applicable specification sheet and any other requirements of this specification not involving tests. Dimensional examination shall be performed to determine conformance to the applicable specification sheet (see 4.3).

4.3.1.1.2 Hose assembly. Each hose assembly shall be visually examined to determine conformance to the applicable specification sheet and any other requirements of this specification not involving tests. Dimensional examination shall be performed to determine conformance to the applicable specification sheet (see 4.3).

4.3.1.1.3 Adapters. Each adapter shall be visually examined to determine conformance to the applicable specification sheet and any other requirement of this specification not involving tests. Dimensional examination shall be performed to determine conformance to the applicable specification sheet (see 4.3).

4.3.1.2 Tests.

4.3.1.2.1 Vent and test valves.

4.3.1.2.1.1 Proof pressure (uncapped valve). The valve, with the cap removed, shall withstand a proof pressure of 465 bars (6,750 lb/in^2) applied to the inlet with the outlet port blocked without evidence of permanent deformation, malfunction, or leakage other than a slight wetting insufficient to form a drop. The proof pressure shall be applied once.

4.3.1.2.1.2 Leakage (uncapped valve). The valve shall be uncapped, actuated, and then subjected to a pressure of 310 bars (4,500 lb/in^2) with no leakage other than a slight wetting insufficient to form a drop.

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4.3.1.2.1.3 Leakage (capped valve). The male connection of the valve cap and probe seal in the valve (see 3.4.1.3) shall be subjected to a pressure of 310 bars (4,500 lb/in^2) with no leakage other than a slight wetting insufficient to form a drop. The method to accomplish this test shall be determined by the manufacturer and may include coupling the cap to a separate valve, with the check valve device removed.

4.3.1.2.2 Hose assembly. The hose assembly shall withstand proof pressure of 620 bars (9,000 lb/in^2) without evidence of permanent deformation or external leakage. Proof pressure shall be applied once.

4.3.1.2.3 Adapters. The adapter shall withstand proof pressure of 465 bars (6,750 lb/in^2) applied to the inlet with the outlet port blocked without evidence of permanent deformation or rupture.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vent and test valves, hose assemblies, and adapters covered by this specification are intended for use in hydraulic systems at a maximum operating pressure of 310 bars (4,500 lb/in^2). Vent and test valves are intended for permanent or temporary installation; whereas, hose assemblies and adapters are intended for temporary test equipment. These components are intended to provide a convenient means of connecting hydraulic system piping or components to instrumentation or gages for the purpose of pressure measurement with a minimum of system fluid loss or contamination. They are also intended for the purpose of taking fluid samples, bleeding air from a hydraulic system, or venting of system pressure with a minimum of safety risk to the operator. Selected adapters are intended for coupling to MIL-DTL-81940/1-1 and MIL-DTL-81940/4-3 sampling and bleed valves to support the above functions. The vent and test valves are intended for use in systems that require two barriers for pressure containment. Material selection for valve should consider compatibility with mating system to avoid galling of valve threads, for example: selection of nickel-copper (NiCu) alloy valve or the use of dry film lubricant in accordance with SAE AS1701 for valve thread coating for an application that involves mating with corrosion resistant steel (CRES) component or piping fitting threads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet.

c. Specification sheet part number (see 1.2).

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24695 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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6.3.1 Assembly drawings. Before test authorization (preferably together with application for qualification and during the examination portion of product qualification), the manufacturer should submit two sets of assembly drawings for each vent and test valve, hose assembly, or adapter for which approval to conduct qualification tests is requested. The following information should be furnished on, or together with, the assembly drawings:

a. Outline dimensions of the complete assembly.

b. Dimensional location of ports, and port sizes.

c. Bill of material, listing specifications, grade, and condition, or other data needed to identify the material. Part numbers should be provided for maintenance replaceable parts.

Dimensioning and tolerancing on drawings should be in accordance with ASME Y14.5.

6.3.2 Source for SD-6 copy. Copies of SD-6, Provisions Governing Qualification, may be obtained from https://quicksearch.dla.mil/or https://assist.dla.mil.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Barrier, pressure, secondary
Bleed, air
Cleanliness, fluid
Contamination
Probe
Sample, fluid

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4820-2023-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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